Exercises for Section 9.3

Directions: For exercises 1 - 4, solve the system of nonlinear equations using substitution.

1. $$\begin{aligned} y &= x-3 \\ x^2+y^2 &= 9 \end{aligned}$$

2. $$\begin{aligned} y &= x \\ x^2+y^2 &= 9 \end{aligned}$$

3. $$\begin{aligned} y &= -x \\ x^2+y^2 &= 9 \end{aligned}$$

4. $$\begin{aligned} x &= 2 \\ x^2-y^2 &= 9 \end{aligned}$$

Directions For exercises 5 - 8, solve the system of nonlinear equations using elimination.

5. $$\begin{aligned} 4x^2-9y^2 &= 36 \\ 4x^2+9y^2 &= 36 \end{aligned}$$

6. $$\begin{aligned} x^2+y^2 &= 25 \\ x^2-y^2 &= 1 \end{aligned}$$

7. $$\begin{aligned} 2x^2+4y^2 &= 4 \\ 2x^2-4y^2 &= 25x-10 \end{aligned}$$

8. $$\begin{aligned} y^2-x^2 &= 9 \\ 3x^2+2y^2 &= 8 \end{aligned}$$

Directions: For exercises 9 - 26, use any method to solve the system of nonlinear equations.

9. $$\begin{aligned} -2x^2+y &= -5 \\ 6x-y &= 9 \end{aligned}$$

10. $$\begin{aligned} -x^2+y &= 2 \\ -x+y &= 2 \end{aligned}$$

11. $$\begin{aligned} x^2+y^2 &= 1 \\ y &= 20x^2-1 \end{aligned}$$

12. $$\begin{aligned} 2x^3-x^2 &= y \\ y &= \frac{1}{2}-x \end{aligned}$$

13. $$\begin{aligned} 9x^2+25y^2 &= 225 \\ (x-6)^2+y^2 &= 1 \end{aligned}$$

14. $$\begin{aligned} x^4-x^2 &= y \\ x^2+y &= 0 \end{aligned}$$

15. $$\begin{aligned} 2x^3-x^2 &= y \\ x^2+y &= 0 \end{aligned}$$

16. $x^2 + y^2 = 9$
$y = 3 - x^2$

17. $x^2 - y^2 = 9$
$x = 3$

18. $x^2 - y^2 = 9$
$y = 3$

19. $x^2 - y^2 = 9$
$x - y = 0$

20. $-x^2 + y = 2$
$-4x + y = -1$

21. $-x^2 + y = 2$
$2y = -x$

22. $x^2 + y^2 = 25$
$x^2 - y^2 = 36$

23. $x^2 + y^2 = 1$
$y^2 = x^2$

24. $16x^2 - 9y^2 + 144 = 0$
$y^2 + x^2 = 16$

25. $3x^2 - y^2 = 12$
$(x-1)^2 + y^2 = 1$

26. $3x^2 - y^2 = 12$
$x^2 + y^2 = 16$